

Design and Thermo-mechanical analysis of CERN's Super Proton Synchrotron internal beam dump

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This Project Report is carried out as a self-study assessment to further develop FEA skills regarding CERN's engineering design challenges, as well as learning more about how proton beams affect beam target's mechanical behaviour under operational loads.

In this document, a detailed thermo-mechanical analysis of CERN's TIDVG5 beam dump is carried out following the reference (1). Its geometry, material properties, thermal contact conductance values, and validation temperatures are directly extracted from Romero Francia et al. (1). A simplified heat source model obtained with Python is used, compared to FLUKA results published in the paper.

1. INTRODUCTION

The Large Hadron Collider Injectors Upgrade (LIU) and High Luminosity LHC (HL-LHC) era presents bigger and more demanding challenges for current devices installed at CERN's facilities. In this document, a further detailed analysis of the TIDVG5 (Target Internal Dump Vertical Graphite, version 5) is carried out following the published paper (1).

Beam dumps are essential systems meant to dissipate beam energy through heat. The TIDVG5 is installed to dissipate proton beam energy from the SPS whenever it is needed, such as in emergency cases, during shutdowns or machine maintenance. Beams are deflected to the absorbing blocks by a set of three vertical kicker magnets (MKDVs) and swept horizontally by means of three horizontal kicker magnets (MKDHs), creating a pattern that dilutes the energy deposited in the dump (1) (see Fig. 1). In the following sections the design and analysis processes are explained.

For more detailed information regarding CERN's SPS TIDVG5 dump refer to the paper used for reference in this document or CERN's official sources.

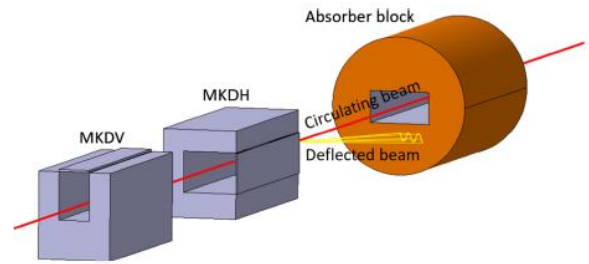


Figure 1. Principle of the main beam dump in the SPS (1)

2. OBJECTIVES

The objectives of this document are to validate the thermo-mechanical response of the TIDVG5 beam dump core using FEA (ANSYS), implement a Python code to obtain the heat deposition model and optimise the beam dump's design through iterative parameter modifications.

3. PROBLEM DEFINITION

Beam energy

The beam parameters used for the analysis of the TIDVG5 are based on CERN's SPS accelerator experiments. Specific details are obtained from the reference (1) and shown in Table 1.

Internal Beam Dump Design for CERN's Super Proton Synchrotron

Table 1. Proton Beam parameters for the TIDVG5 beam dump

Parameter	Nominal (LHC)	Worst-Case (FT)
Beam Momentum	14–450 GeV/c	14–450 GeV/c
Avg. Beam Power	164 kW	270 kW
Pulse Energy	5.60 MJ / 2.88 MJ	2.88 MJ
Cycle Period	36 s	10.8 s

Beam dump core design

The TIDVG5's absorbing core is formed of solid blocks of 4.4 m graphite, 0.2 m of TZM (Titanium Zirconium Molybdenum) and 0.389 m of W (Tungsten) (see Fig. 2). These blocks are enclosed by several CuCr1Zr cooling plates and a 5-meter multidirectionally forged 304L SS vacuum chamber (1).

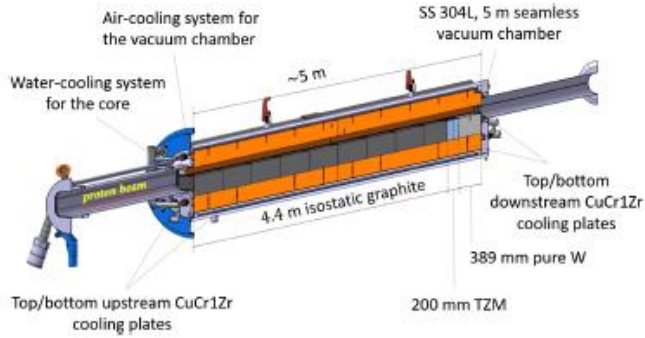


Figure 2. Cross section of the TIDVG5 SPS internal dump core, the cast-iron first shielding is hidden for clarity (1)

A basic re-interpretation of the target design is assembled in CAD software Inventor Professional, as shown in Fig. 3, where the detailed model of the core is included and implemented for further analysis.

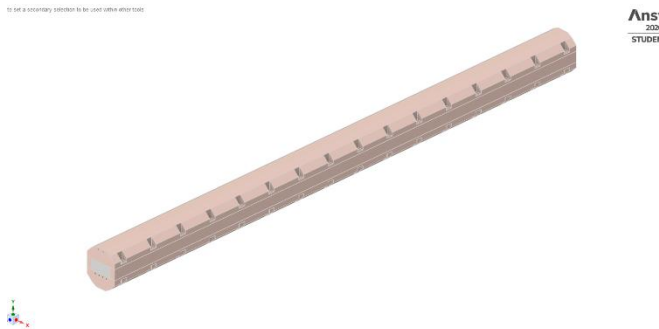


Figure 3. CAD assembly of the studied internal beam dump

Some minor changes were made in the geometry of the beam core in order to simplify the model and better understand how parameter changes affect the results.

In the following table, dimensions and exact numbers of blocks of the studied design is shown,

Material	Blocks	Length (m)
Graphite	9	$8 \times 0.5 + 1 \times 0.4$
TZM	2	0.1
W	1	0.389

TIDVG5 Core material properties

Material properties are taken from the reference paper mentioned along the document. They are shown in Table 3 and adapted to the studied geometry.

Table 2. Material properties

Material	Density (g/cm ³)	Number of blocks	Block length (cm)
Graphite R7550	1.8	9	440
TZM	10.1	2	20
Tungsten	18.8	1	39
CuCr1Zr	8.9	1	1

4. METHODOLOGY

Software

The software used to study the TIDVG5's beam dump core design are:

- Inventor Professional 2026: CAD geometry
- Python: beam heat deposition model
- ANSYS: Thermo-mechanical FEA

Heat Deposition Model

Monte Carlo Simulations (FLUKA) are used for calculating energy deposition profiles. This code simulates the particle transport and provides a detailed 3D map of the energy deposited.

In this project, an alternative to FLUKA was developed using Python to simulate energy deposition. The model employs a Parametrised Hadronic Shower approach. It uses a Gamma distribution to represent the longitudinal energy profile and a Gaussian distribution to model the transverse spread. This methodology captures the fundamental physics of a 450 GeV proton beam interacting with the absorber material, with some limitations which will be later discussed, while significantly reducing processing time.

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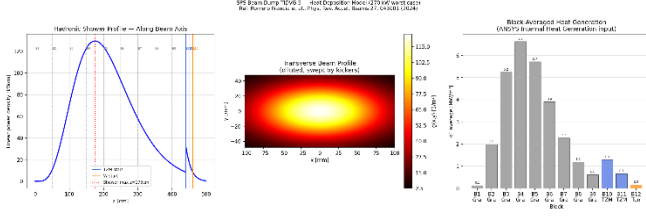


Figure 4. Heat Deposition Model – 270 kW worst case scenario

Thermo-mechanical analysis – 270 kW beam power

A thermo-mechanical finite-element analysis of the absorbing blocks is carried out with ANSYS to study its response in the worst-case operation scenarios for beam dumping.

The simulation workflow is shown below. A Steady-State Thermal analysis is performed to obtain the temperatures and heat flux along the beam dump core. These results are then imported to a Static Structural analysis to assess stress and deformations due to the material's heat expansion.

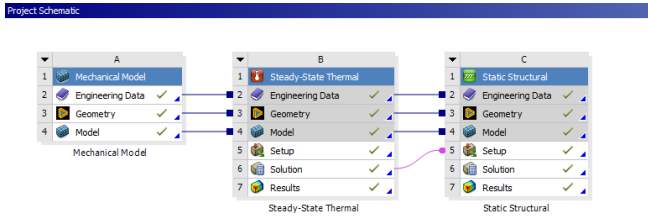


Figure 5. Project workflow in ANSYS Workbench

A. Material assignment

Material properties are introduced in ANSYS through the Engineering Data menu. When the geometry has been successfully imported as the Mechanical model, the absorbing blocks are given its corresponding material, shown with specific colours (Fig. 6).

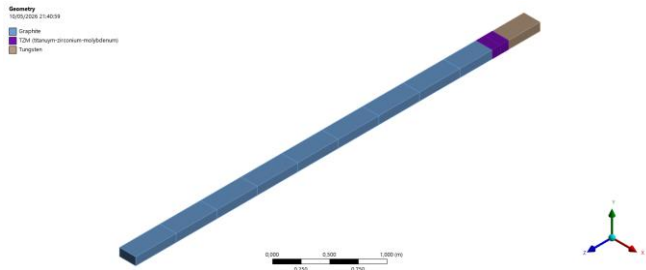


Figure 6. Material assignment for each block in ANSYS Mechanical

B. Connections

Bonded contact connections are used between the absorbing block end faces.

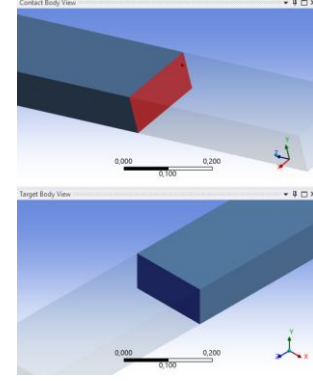


Figure 7. Bonded connections of the absorbing blocks

C. Mesh

Hexahedral elements are preferred as they provide better accuracy and a faster rate of convergence than other mesh element types (2).

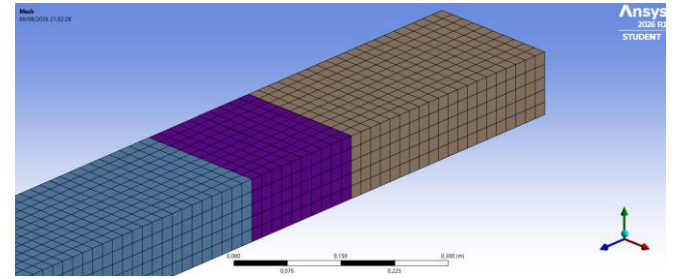


Figure 8. Mesh model in ANSYS

Mesh Quality Worksheet				
Item	Quality	Warning Count	Error (Default) Count	Value
Hex Element Quality	Default (0.05)	Default (0.05)	Default (0.05)	0.05
Hex Element Ratio (Corner Nodes)	Default (0.05)	Default (0.05)	Default (0.05)	0.05
Hex Element Ratio (Corner Points)	Default (0.05)	Default (0.05)	Default (0.05)	0.05
Hex Element Edge Length	2.000 m	4.000 m	4.000 m	0.001 m
Hex Element Angle	Default (120 °)	Default (120 °)	Default (120 °)	90 °
Hex Element Edge Length	0.001 m	2.0e-05 m	0.001 m	0.001 m
Hex Element	Default (0.05)	Default (0.05)	Default (0.05)	0.05
Hex Tet Volume	Default (0.05)	Default (0.05)	Default (0.05)	0.05
Hex Skinning Angle	Default (10 °)	Default (10 °)	Default (10 °)	5.0e-01 °

Figure 9. Mesh quality worksheet in ANSYS

D. Boundary Conditions

As mentioned earlier, the Python script gives the volumetric heat generation q''' (W/m³) per absorbing block, which are introduced in ANSYS as “Internal Heat Generation”.

Thermal Contact Conductance (TCC) values between system components are directly taken from the reference paper (1).

Table 3. TCC values used for thermo-mechanical studies

	TCC (W/(m ² K))
Graphite-CuCr1Zr cooling plates	900
TZM-CuCr1Zr cooling plates	900
W-CuCr1Zr cooling plates	800

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E. Mesh Convergence

Mesh convergence was assessed by comparing peak graphite temperatures across three mesh densities. The medium mesh (25 mm, 34652 nodes) produces results within less than 1,0% of the fine mesh (20 mm) solution and is adopted for all subsequent analyses.

F. Steady-State Thermal Analysis

Temperature field across the absorbing blocks is shown in Figure 9 for the worst-case operation scenario, which corresponds to a thermal beam power of 270 kW. Peak temperatures of 449 °C occur at block number 4, which is expected as shown in the reference paper (1). This confirms that the heat deposition model correctly predicts the maximum temperature location within the graphite blocks.

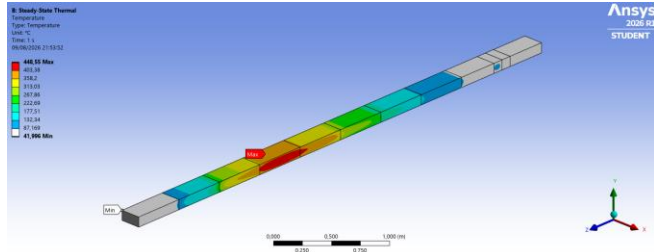


Figure 10. Temperature (°C) results in the absorbing blocks

On the TZM blocks, however, a peak temperature of 89 °C is obtained in comparison with the 505 °C that the reference paper predicted. This discrepancy is most likely to be due to the absence of full physics involved in the heat deposition model. FLUKA simulation captures neutron and charged pion energy deposition in high density materials, which result in heat generation levels that surpass those from just the primary shower profile.

Total Heat Flux is assessed to ensure the block's faces correctly dissipate heat as established through the boundary conditions. As observed in the following figure, the top and bottom faces of the absorbing blocks simulate correctly the heat dissipation towards the CuCr1Zr cooling plates, especially where peak temperatures are obtained.

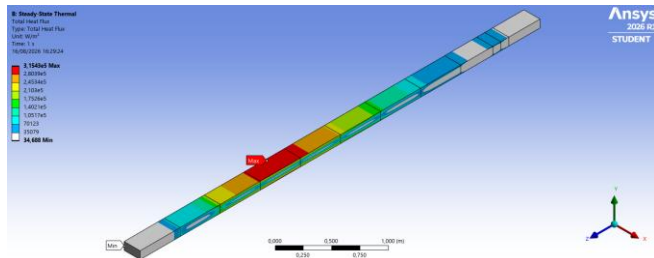


Figure 11. Total Heat Flux (W/m²) results in the absorbing blocks

Global Energy Balance is verified to ensure temperature fields and other results are correct. This is done by checking the “Reaction heat probe” across the blocks.

$$Q_{input} = +270 \text{ kW} ; Q_{reaction} = -269,4 \text{ kW}$$

A difference of 0,2% is achieved.

Thermal Results -Validation

Table 4. Peak temperatures obtained in the FEA and compared to the reference paper (1).

Material	Blocks	Max. Temp. FEM (°C)	Max. Temp. (°C) Ref. Paper	Difference (%)
Graphite	1 to 9	449	532	-15.6
TZM	10 to 11	89	505	-82.4
Tungsten	12	64	98	-34.7

The maximum temperature obtained at the graphite block number 4, with a value of 449 °C and a 15.6% difference compared to the reference paper (1). This is considered acceptable as the heat deposition model does not fully interpret the physics as FLUKA simulations do.

On the other hand, maximum temperatures at the TZM and W absorbing block represent a major discrepancy, especially TZM with a 82,4% difference. This may be due to the uniform heat generation input in ANSYS and, as mentioned earlier, the alternative and simplified use of a heat deposition model using Python.

Table 5. Christensen and von Mises Criterion

Block	Max Stress - FEM (MPa)	Criterion < 1
Graphite (Christensen)	17.43 (Compression)	0.13
TZM (von Mises)	107.8	0.16
Tungsten (Christensen)	172.5 (Compression)	0.10

The Christensen and von Mises criterion of the different materials are studied. TZM is assessed under the von Mises criterion due to its ductile nature (3). On the other hand, both Graphite and Tungsten are brittle material and the Christensen criterion is used.

All materials are safe and no failure is predicted during the particle beam power of 270 kW.

6. CONCLUSIONS

A thermomechanical study of CERN's SPS TIDVG5 has been carried out to validate its design against higher intensity and more powerful beams. This assessment is necessary as the internal beam dump is required to withstand during the HL-LHC operations.

The model was referenced and validated using available papers, more specifically, the reference (1). A simplified and faster heat deposition model based on hadronic shower theory was developed in Python as a substitute of FLUKA simulations. This resulted in lower peak temperatures across the absorbing blocks, which corresponded in lower stress levels as the materials don't suffer the same amounts of thermal expansions. This indicates a consistent and reliable simulation model of the TIDVG5 to which further work can go into.

In future studies a more complex heat deposition model will improve the analysis as it will provide with more accurate heat generation values across the blocks.

7.0 REFERENCES

1. **al., A. ROMERO FRANCIA et.** *Design and early operation of a new-generation internal beam dump.* Geneva : CERN, 2024.
2. **Tadepalli, S. C., Erdemir, A., & Cavanagh.** *Comparison of hexahedral and tetrahedral elements in finite element analysis of the foot and footwear.* 2011.
3. **Mises, R. von.** *Mechanics of solid bodies in the plastically-deformable state.* 1913.
4. **Christensen, Richard M.** *A Comprehensive Theory of Yielding and Failure for Isotropic Materials.* 2007.